

Mini Literature Review: Smart Parking System with Vehicle Detection and Barrier Control

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I. PROBLEM STATEMENT

Project Topic: Smart Parking System with Vehicle Detection and Barrier Control.

Traditional parking systems in campuses, offices, and residential areas still rely heavily on manual operation. Vehicle entry and exit are often handled without automatic verification, centralized monitoring, or event recording. This causes slow processing, inaccurate parking records, and limited ability to review parking history.

Sensor-only parking systems are low-cost, but they cannot verify whether a detected object is actually a valid vehicle. Therefore, this project proposes a hybrid IoT-based smart parking prototype that combines ultrasonic sensing, ESP32-CAM image capture, backend AI processing, barrier control, and dashboard monitoring.

Objective: Develop a low-cost prototype that can detect vehicle presence, capture vehicle images, control the barrier, store parking events, and display system status through a web dashboard.

II. SEARCH STRATEGY

The literature search was conducted using IEEE Xplore, Scopus, Google Scholar, publisher websites, and arXiv. The keywords were: “Smart Parking System AND IoT”, “Ultrasonic Sensor AND Smart Parking”, “ESP32-CAM AND Parking System”, “Vehicle Access Control AND Smart Gate”, and “YOLO Vehicle Detection AND Parking”.

More than ten candidate papers were found. Seven papers were selected because they support at least one design component of this project: ultrasonic sensing, ESP32-CAM image capture, vehicle detection, barrier control, backend storage, or dashboard monitoring. Papers were excluded when they were paid, outdated, weakly related, or lacked implementation details.

III. LITERATURE SUMMARY

TABLE I
LITERATURE SUMMARY OF SELECTED PAPERS

Paper	Problem	Solution	Technology	Metrics	Limitation
P1 [1]	Lack of real-time parking data	IoT monitoring system	Arduino, camera, web and mobile app	Real-time status	Less focus on gate control
P2 [2]	Need parking availability updates	Smart parking management	Arduino, ultrasonic sensor, MQTT, database	Status update	No camera verification
P3 [3]	Parking search wastes time	IoT sensor prototype	IoT sensors and controller	100% accuracy	Cannot verify valid vehicle
P4 [4]	Need simple slot detection	Ultrasonic parking system	Arduino, ultrasonic sensor, IR sensor, Wi-Fi	Slot detection	No image record
P5 [5]	Need contactless parking portal	ESP32-CAM gate system	Arduino UNO, HC-SR04, ESP32-CAM, servo, LCD	100% test result	Limited event history
P6 [6]	Need vehicle identification	Plate detection monitoring	ESP32, sensors, servo, web camera	Plate monitoring	Depends on image quality
P7 [7]	Sensor-only detection is limited	YOLO vehicle detection	YOLOv8–YOLOv11	99.68% balanced accuracy	High computation cost

IV. COMPARATIVE ANALYSIS

TABLE II
TECHNOLOGY COMPARISON

Paper	Sensor	Camera	AI/CV	Barrier	Dashboard
P1	Y	Y	P	–	Y
P2	Y	–	–	–	P
P3	Y	–	–	–	P
P4	Y	–	–	P	–
P5	Y	Y	P	Y	P
P6	Y	Y	Y	P	Y
P7	–	Y	Y	–	–

The comparison shows four main trade-offs. First, ultrasonic sensors are fast, cheap, and simple to integrate, but they only detect object presence and cannot verify vehicle validity. Second, camera-based systems provide visual evidence and future license plate recognition, but they increase cost and processing complexity. Third, YOLO-based detection improves vehicle verification accuracy, but backend processing is more practical than running AI directly on ESP32-CAM. Fourth, dashboard monitoring improves centralized management, but many systems do not fully connect dashboard records with physical gate decisions.

Based on these findings, the proposed system uses a hybrid architecture: ultrasonic sensors provide the first trigger, ESP32-CAM provides visual evidence, the backend server performs AI verification, and the dashboard stores and displays parking events.

V. RESEARCH GAP

Three measurable gaps are identified from the literature. First, many low-cost systems evaluate ultrasonic detection but do not measure vehicle verification accuracy after camera confirmation. Second, existing studies rarely report end-to-end delay from vehicle detection to final barrier response. Third, several systems provide either detection or monitoring, but fewer systems integrate sensor readings, images, AI results, gate state, and parking history into one dashboard.

VI. RESEARCH QUESTIONS

TABLE III
RESEARCH QUESTIONS AND METRICS

RQ	Research Question	Metric
RQ1	How accurately can the system detect and verify vehicle presence near the parking gate?	Detection Accuracy (%)
RQ2	How long does the system take from vehicle detection to barrier response?	Response Delay (ms)

VII. DESIGN DECISION

TABLE IV
DESIGN DECISION

Component	Alternatives	Selected	LR	Technical Reason
MCU	Arduino, ESP32, NodeMCU	Arduino UNO + ESP32-CAM	P3,P5	Separate stable gate control from image capture and Wi-Fi upload
Detection	Sensor only, camera only, hybrid	Hybrid	P3,P5,P7	Reduce false triggers by combining fast sensing with visual verification
Communication	HTTP, MQTT, Web-Socket	HTTP	P1,P5	Simple for ESP32-CAM image upload and FastAPI backend integration
AI Processing	On-device, edge, backend	Backend AI	P7	YOLO requires more computing resources than ESP32-CAM can provide
Platform	LCD, mobile app, dashboard	Web dashboard	P1,P6	Supports centralized monitoring, images, and parking event history

VIII. PROPOSED BLOCK DIAGRAM

Figure 1 shows the proposed smart parking architecture. The system includes Entry and Exit Sensor-Camera Modules, an Arduino Master Controller, a backend AI server, an SQL database, and a web dashboard.

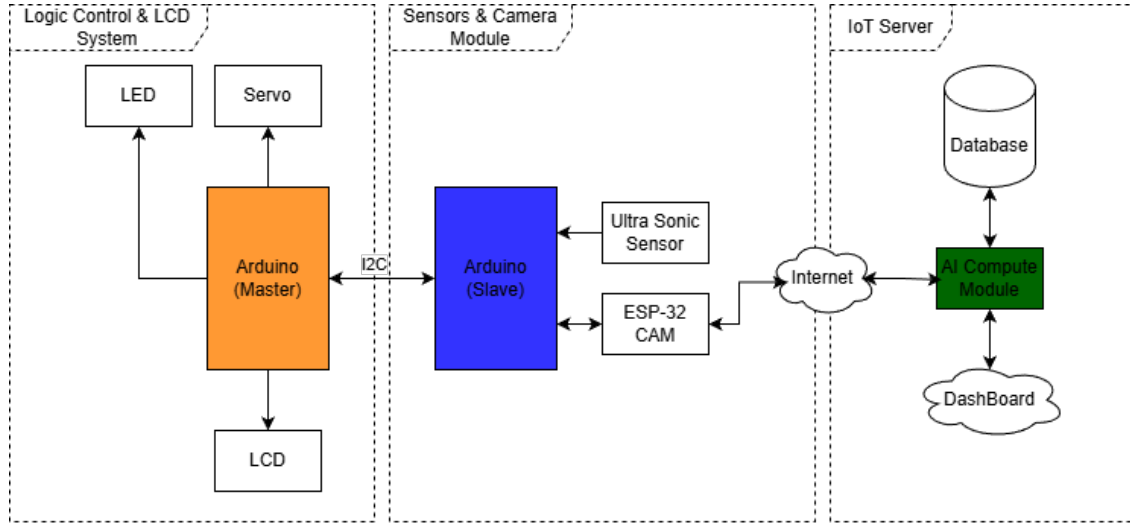


Fig. 1. Proposed Smart Parking Architecture

TABLE V
MAPPING FROM LITERATURE REVIEW TO BLOCK DIAGRAM

Block Component	LR Evidence	Reason
Ultrasonic Sensor	P3,P4	Fast and low-cost vehicle-presence detection
ESP32-CAM	P5,P6	Image capture for vehicle verification and record keeping
Arduino Master	P5	Local barrier, LCD, and indicator control
YOLO Backend Server	P7	Vehicle detection with stronger computing resources
SQL Database and Dashboard	P1,P6	Centralized storage, monitoring, and history review

IX. EXPECTED METRICS

TABLE VI
EXPECTED METRICS

Metric	Unit	Target
Vehicle Detection Accuracy	%	$\geq 95\%$
Gate Response Delay	ms	≤ 3000
Dashboard Update Delay	ms	≤ 1000
System Uptime	%	$\geq 95\%$

These metrics are selected because they directly evaluate vehicle detection reliability, barrier response speed, dashboard monitoring delay, and overall prototype stability.

X. AI REFLECTION

TABLE VII
AI AUDIT LOG

Prompt	AI Output	Verification	Reflection
Suggest smart parking papers using ESP32-CAM and IoT	Suggested several candidate papers and related technologies	Checked paper availability and relevance manually	Only accessible and relevant papers were kept
Compare ultrasonic detection, camera verification, and YOLO	Recommended a hybrid detection architecture	Compared with selected papers P3, P5, P6, and P7	Hybrid architecture was selected instead of sensor-only detection
Recommend evaluation metrics for smart parking	Suggested accuracy, delay, and uptime	Compared with research gaps and project objectives	Dashboard update delay was added as a project-specific metric

AI tools were used to support paper exploration, technology comparison, and metric selection. All paper information, technical claims, and design decisions were manually verified before being included in this review.

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